Ruby Neisser - Mar 18, 2025, 9:48 AM CDT

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cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ nano ch2oh-_Geom_sym1.inp
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ ./xit -z <ch2oh-_Geom_sym1.lo
g> ch2oh-_Geom_sym1.out
STOP CAN'T IDENTIFY RUN TYPE.
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ rm ch2oh-_Geom_sym1.inp
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ nano ch2oh-_Geom_sym1.log
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ ./xit -z <ch2oh-_Geom_sym1.lo
g> ch2oh-_Geom_sym1.out
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ nano ch2oh-_Geom_sym1.out
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ ls
CO.xyz                                   ch2oh-_water1f.gjf
CO_H2O_1.xyz                             ch2oh-_water1f.log
Ediff.f                                  ch2ohrad_f1.log
H2O.xyz                                  ch2ohrad_f2.log
H2O_opt.inp                              co+h2o_1.rxn
PracticeCalculations.pdf                 h
Quantum-Chemistry-for-Ices_Woon_v01.pdf  hydroxymethyl_radical_Geom_1.log
ch2oh-_Geom_sym1.log                     hydroxymethyl_radical_Geom_2.log
ch2oh-_Geom_sym1.out                     mull.f
ch2oh-_freq.log                          phasit.f
ch2oh-_water1.gjf                        temp.alias
ch2oh-_water1.log                        xit
ch2oh-_water1.xyz                        xit.f
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ nano ch2oh-_freq.log
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ ./xit -z < ch2oh-_freq.log >
ch2oh-_freq.out
cewalker6@DESKTOP-K3BHET1:~/Dave_Woon_Results$ nano ch2oh-_freq.out
```

Gfortran (install in wsl first using sudo apt install gfortran)

- Copy all files (.f, .log, .xyz, .rxn) into ubuntu into the same directory, and make sure you cd to that directory.
- Compile fortran code into executables:
  - input/ $ gfortran -o myprogram myprogram.f
- input/ $ gfortran -o xit xit.f
  - output/
    - xit.f:180:27:

```
  180 |           close(unit=10,dispose="delete")
      |                          1
Error: Syntax error in CLOSE statement at (1)
xit.f:272:25:

  272 |         close(unit=10,dispose="delete")
      |                        1
Error: Syntax error in CLOSE statement at (1)
```

  - Replace "dispose" with "status" in xit.f, then rerunning should compile correctly
- Running executables: $ ./myprogram
  - ./ may not be necessary depending on your environment
- input/ $ ./xit -z < optorfreqoutput.log > nameforoutputtobewrittento.out
  - output/ STOP CAN'T IDENTIFY RUN TYPE.
  - xit is specifically looking for the Gaussian keywords to start with " # B3LYP/basis" or similar; method must come first & all caps, so edit your .log file to match that. Rerunning will work
  - Quiet output, if you open the output file you will see xit has grabbed the cartesian coordinates,

energy, and RMS Force and Displacement at each stage of the optimization in the file. First line is # atoms, then energy and RMS, then the coordinates.

- input/ $ gfortran -o mull mull.f
- input/ $ ./mull < outputcontainingmullikens.log
  - output/ At line 8 of file mull.f (unit = 5, file = 'stdin')
    Fortran runtime error: Bad value during integer read

    Error termination. Backtrace:
    #0 0x7f8edd694e59 in ???
    #1 0x7f8edd695a71 in ???
    #2 0x7f8edd9294d3 in ???
    #3 0x7f8edd92e029 in ???
    #4 0x7f8edd92f5e2 in ???
    #5 0x55fe0c756272 in MAIN__
    #6 0x55fe0c7569fa in main
- input/ $ ./mull < freqoutput.log > test.out
  - Quiet output. test.out contains one line: No data! *To Be Troubleshot
- input/ $ gfortran -o Ediff Ediff.f
- Before using Ediff, create .xyz and .rxn files following the CO + H2O examples (ZPE in kcal/mol). Don't need to remake H2O.xyz, just use Dave's. You can grab the EB3LYP and xyz coordinates from the xit output of the frequency file, though you might need to round EB3LYP to lower precision (check CO.xyz). Copy the ZPE from your frequency output file.
- input/ $ ./Ediff < co+h2o_1.rxn
  - output/ ---
    ```
    BIMOLECULAR REFERENCE : asymptote
                                   E(Eh)     ZPE(kcal/mol)
    CO.xyz                      -113.330281      3.125
    H2O.xyz                      -76.443387     13.337
    NET                         -189.773668     16.462
    ---------------------------------------------------------------------
    DIFFERENCE  1 of  1 : complex
    CO_H2O_1.xyz                      -189.776968      17.500
    NET                         -189.776968     17.500
    Delta Ee       -2.071  kcal/mol
    Delta E0       -1.033  kcal/mol
    ---------------------------------------------------------------------
    ```
- input / $ ./Ediff < ch2oh-_water1.rxn > outputfile.ext
  - Quiet output, writes output to the file